A Modeling Framework for Scalable Near-Duplicate Detection Using Random Projections and Locality-Sensitive Hashing

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Abstract

High-dimensional datasets pose a fundamental challenge for similarity-based retrieval systems and clustering due to the curse of dimensionality and the very high computational costs associated with exact search methods. In this project, we present a modelling-driven study of approximate similarity preservation, grounded in the Johnson–Lindenstrauss (JL) Lemma, and its practical application in Locality-Sensitive Hashing (LSH). Data objects are modelled as points in a high-dimensional metric space, and the similarity metric is formalised as a distance-preserving geometric property. The JL Lemma is introduced as a probabilistic-modelling guarantee, proving that random linear projections can approximately preserve pairwise distances within bounded distortion, enabling efficient dimensionality reduction without significant loss of structural information. LSH is employed as a modelling framework that maps geometric closeness to collision probability, allowing sublinear-time approximate nearest neighbour search. We perform an explicit analysis of the model's assumptions, parameters, and trade-offs including distortion tolerance, false positive/negative balance, and collision rates. The framework is validated using a real-world application of near-duplicate text detection on publicly available document datasets, and experimental results compare LSH-based similarity retrieval against established clustering frameworks such as K-Means and HDBSCAN across multiple data scales. The results showcase how existing clustering frameworks often struggle with scalability and are highly sensitive to increased dimensionality, while the JL–LSH model achieves orders-of-magnitude speedups with superior retrieval quality. The proposed framework has broader applications in plagiarism detection, content-based recommendation systems, web-scale document deduplication, bioinformatics sequence matching, and anomaly clustering in cybersecurity pipelines.

Key words. Johnson–Lindenstrauss Lemma, Locality-Sensitive Hashing, random projections, near-duplicate detection, approximate nearest neighbour search, dimensionality reduction, multi-modal similarity, graph-based clustering.

Contents

1. Introduction

In every deep learning (DL) domain—such as natural language processing (NLP), computer vision (CV), and speech processing—there has been a rapid growth of high-dimensional datasets. This has led to similarity search and near-duplicate detection becoming a fundamental problem in modern machine learning (ML) and deep learning systems. Given a dataset of objects embedded in a metric space $(\mathcal{X}, d)$, the goal is to efficiently identify points that are either exact or approximate nearest neighbours. The classical approach to nearest-neighbour search suffers from the *curse of dimensionality*, wherein computational complexity grows exponentially as the dimension d increases, making brute-force search methods impractical for most large-scale scenarios.

To address these challenges, two powerful theoretical frameworks have been developed: the Johnson–Lindenstrauss (JL) Lemma [1] and Locality-Sensitive Hashing (LSH) [2]. LSH is built upon the foundation of the JL Lemma [1] to preserve pairwise distances while reducing dimensions. Together, they provide a novel approach for dimensionality reduction and similarity search in sublinear time complexity [3], forming a strong foundation for scalable near-duplicate detection systems.

Johnson–Lindenstrauss Lemma: Geometry of Dimension Reduction. The JL Lemma [1] states that any set of n high-dimensional points can be embedded into a much lower-dimensional space while approximately preserving the pairwise distances between all points. This foundational result is stated formally below.

Lemma 1.1 (Johnson–Lindenstrauss [1])**.** *For any $0 < \varepsilon < 1$ and any set of n points in $\mathbb{R}^d$, there exists a mapping*

$$f : \mathbb{R}^d \to \mathbb{R}^k$$

such that for all points x_i, x_j,

$$(1-\varepsilon)\|x_i - x_j\|_2^2 \;\leq\; \|f(x_i) - f(x_j)\|_2^2 \;\leq\; (1+\varepsilon)\|x_i - x_j\|_2^2,$$

where the target dimension satisfies

$$k = O\left(\frac{\log n}{\varepsilon^2}\right).$$

This result implies that high-dimensional geometry is intrinsically low-dimensional, paving the way for efficient processing without significant distortion of pairwise structure. The proof relies on probabilistic constructions using random Gaussian projections, exploiting the concentration-of-measure phenomenon on high-dimensional spheres [1, 5]. In our experimental pipelines, high-dimensional feature vectors—TF-IDF representations for text, log-mel features for speech, and pixel features for images—are each mapped into a reduced space using this principle, resulting in low distortions in cosine distances across all modalities.

Approximate Nearest Neighbour Search and LSH. While the JL Lemma [1] reduces dimensionality, efficient similarity search still requires sub-quadratic comparisons. This challenge is resolved by Locality-Sensitive Hashing (LSH), introduced by Indyk and Motwani [2], a probabilistic framework that ensures similar points collide with higher probability than dissimilar ones. The combination of JL projections and LSH hashing was shown to yield an efficient approximate nearest neighbour (ANN) algorithm with polynomial-time preprocessing and logarithmic query complexity [2, 3, 6]. In the modern similarity-search pipeline, JL and LSH are complementary: JL provides a compact representation that preserves structure, while LSH leverages this representation for fast candidate retrieval [3, 6].

Multimodal Near-Duplicate Detection. A key challenge in modern deep learning applications is that data is inherently multimodal, spanning text, images, and audio. Each modality carries its own feature representation: NLP documents use TF-IDF or embedding vectors, computer vision data uses flattened pixel vectors or learned features, and speech data uses log-mel spectrogram representations. Despite this heterogeneity, all modalities can be unified under a vector similarity framework, enabling the application of the JL–LSH pipeline [1, 2] uniformly across domains. Our framework validates this unified approach across six toy datasets (text, audio, and images) and one large-scale Wikipedia experiment.

Project roadmap. The project unfolds in two broad phases. In the **first phase**, the team establishes the theoretical and algorithmic foundation: the JL Lemma [1] is introduced as a probabilistic guarantee for distance-preserving dimensionality reduction; LSH [2] is formalised as a locality-sensitive hashing framework; and the complete modelling pipeline (feature extraction, JL projection, LSH hashing, cosine refinement, and graph-based clustering) is designed and implemented. In the **second phase**, the framework is validated experimentally. Six toy experiments are conducted across text (20 Newsgroups, AG News), audio (Speech Commands, Spoken Digit), and image (CIFAR-10, Fashion-MNIST) datasets, comparing the JL + LSH approach [1, 2, 3] against KMeans and HDBSCAN on clustering quality and runtime. The phase culminates in a large-scale Wikipedia experiment on 12,000 articles.

Report structure. Section 2 reviews related work and the limitations of classical clustering methods. Section 3 develops the complete methodology, including feature extraction, JL projection, LSH hashing, candidate refinement, and graph-based clustering. Section 4 presents the experimental setup and datasets. Section 5 reports results across toy and large-scale experiments. Section 6 discusses findings. Section 7 concludes.

2. Background and Related Work

2.1. Limitations of Traditional Clustering in High Dimensions

Extensive study in computational geometry, machine learning, and information retrieval has highlighted the limitations of traditional similarity search and clustering in high-dimensional spaces. Classical algorithms such as KMeans, Hierarchical Clustering, and HDBSCAN face four fundamental challenges:

- **Distance Concentration.** In high dimensions, pairwise distances between any two points tend to become nearly equal, effectively reducing the discriminative power of distance-based clustering. This renders centroid- and density-based methods highly unreliable as the ambient dimension grows.
- **Computational Complexity.** Algorithms like KMeans require repeated distance computations across all data points, leading to $O(nkd)$ time complexity per iteration, which becomes infeasible for large n and d.
- **Curse of Dimensionality.** Indexing structures such as KD-trees degrade to near-linear scan performance as dimensionality increases, eliminating their theoretical advantage over brute-force search.
- **Sensitivity to Noise and Sparsity.** High-dimensional data—particularly speech and vision features—are often sparse, destabilising centroid- and density-based clustering methods.

These limitations make classical approaches almost infeasible and unsuitable for the near-duplicate detection task, where the goal is to identify highly similar items across large datasets.

2.2. The Johnson–Lindenstrauss Lemma

The JL Lemma originates from the seminal work of Johnson and Lindenstrauss [1]. It shows that a set of n points in a high-dimensional Euclidean space can be mapped to a much lower-dimensional space of dimension $k = O(\log n/\varepsilon^2)$ while approximately preserving all pairwise distances. The proof relies on probabilistic constructions using random projections through Gaussian or sparse matrices [1, 5]. Random orthogonal projections preserve distances with high probability owing to the concentration-of-measure phenomenon on high-dimensional spheres.

Remark 2.1. In practical implementations, the JL Lemma [1] is applied as a preprocessing step before downstream tasks such as clustering or nearest-neighbour search. Our experimental pipeline converts TF-IDF representations to JL-projected vectors before applying LSH [2, 3], reducing computational complexity while maintaining structural similarity.

2.3. Locality-Sensitive Hashing

A major breakthrough in approximate nearest-neighbour (ANN) search was Locality-Sensitive Hashing (LSH), introduced by Indyk and Motwani [2]. Rather than finding exact neighbours, ANN search aims to find a point within a factor of $(1 + \varepsilon)$ of the optimal distance. LSH is based on the idea of hashing points such that similar points collide with high probability and dissimilar points collide with low probability [2].

Definition 2.2 (Locality-Sensitive Hash Family [2])**.** A hash family $\mathcal{H}$ is called *locality-sensitive* if for any points p, q:

$$\Pr_{h\in\mathcal{H}}[h(p) = h(q)] = \begin{cases} \text{high}, & \text{if } d(p,q) \leq r, \\ \text{low}, & \text{if } d(p,q) \geq (1+\varepsilon)r. \end{cases}$$

This probabilistic separation allows efficient indexing into hash buckets and enables sublinear query time for nearest-neighbour search [2, 6]. The original work demonstrated that combining random projections from the JL Lemma [1] with hashing yields an efficient ANN algorithm with polynomial-time preprocessing and logarithmic query complexity. The subsequent extension by Gionis, Indyk, and Motwani [3] further demonstrated the power of this combination for high-dimensional similarity search via the banding amplification technique.

In the modern similarity-search pipeline, JL and LSH are complementary [3]: JL reduces dimensionality while preserving distances, and LSH enables fast search in the reduced space.

2.4. Our Approach: Combining Two Powerful Frameworks

The limitations surveyed above motivate a two-component strategy that directly targets the bottlenecks of exact similarity search in high dimensions.

1. **Johnson–Lindenstrauss Random Projection [1].** The JL Lemma [1] (Lemma 1.1) compresses each high-dimensional feature vector into a compact k-dimensional representation, where $k = O(\log n/\varepsilon^2)$. Because the projection is a random linear map, it requires no labelled data, no model training, and no knowledge of the downstream task, making it universally applicable across text, audio, and image modalities.

2. **Locality-Sensitive Hashing [2, 3].** LSH [2] is applied in the JL-reduced space to generate candidate similar pairs in sublinear time. Random hyperplane hashing assigns each point a binary signature; points sharing the same signature in at least one hash band are identified as potential near-duplicates [3, 4]. Candidate pairs are subsequently refined with exact cosine similarity to remove false positives, and surviving pairs are organised into near-duplicate clusters via graph-based connected components.

The combined JL + LSH pipeline [1, 2, 3] delivers three key properties that classical clustering methods lack:

- **Modality agnosticism.** Because both JL [1] and LSH [2, 4] operate purely on normalised real-valued vectors, the same pipeline applies unchanged to text (TF-IDF), audio (log-mel spectrograms), and images (pixel intensities) by simply substituting the feature extraction front-end.
- **Scalability.** The overall complexity of $O(ndk + nBk + |C|)$ grows sub-quadratically in the number of data points n, compared to $O(n^2 d)$ for exact cosine search or $O(n^3)$ for brute-force methods.
- **Theoretical guarantees.** The JL Lemma [1] provides a probabilistic bound on distortion, and the LSH framework [2] provides explicit control over false positive and false negative rates through the choice of hash bits B and band size b—giving practitioners interpretable levers over the precision–recall trade-off.

The full mathematical treatment of each component is given in Section 3.

3. Methodology

We present a scalable, unified framework for near-duplicate detection across multimodal datasets—text, speech, and computer vision—using JL projections [1] and LSH [2]. The pipeline is designed to address computational challenges arising from high dimensionality and large-scale corpora while maintaining strong approximation guarantees [3].

3.1. Pipeline Overview

The complete methodology consists of a sequence of transformations applied to raw data, culminating in graph-based clustering of near-duplicate items. The modular structure ensures that each component can be independently analysed, fine-tuned, and reused across modalities.

1. High-dimensional features are extracted and normalised from any given modality.
2. JL random projections [1] reduce dimensionality while preserving cosine similarity.
3. LSH [2] generates candidate clusters via hash collisions.
4. Graph-based clustering (connected components) identifies near-duplicate groups.

Figure 1 provides a high-level schematic of the complete pipeline, showing how raw multimodal inputs flow through each stage to produce near-duplicate clusters and evaluation metrics. Figure 2 presents the detailed step-by-step workflow for each processing stage.

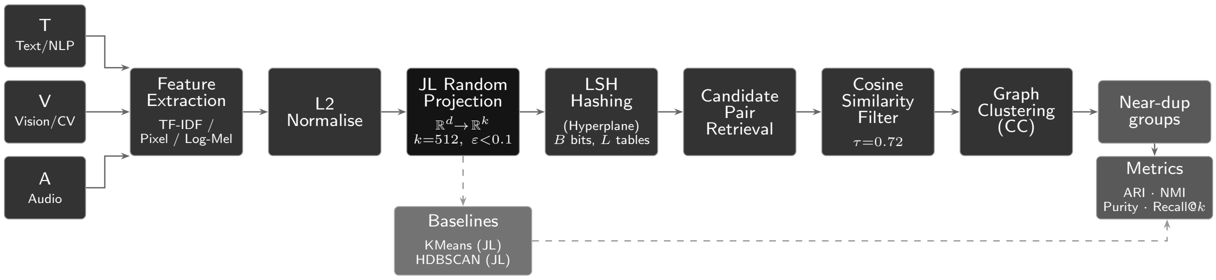

Figure 1: Overview of the proposed scalable multimodal near-duplicate detection pipeline. Raw inputs from text, vision, and audio domains are first transformed into high-dimensional feature representations (TF-IDF, pixel values, log-mel spectrograms) and L2-normalised. These features are then projected into a lower-dimensional space using Johnson–Lindenstrauss (JL) random projections [1] to preserve pairwise similarities. Locality-Sensitive Hashing (LSH) [2] with random hyperplanes generates hash buckets for efficient candidate pair retrieval. Retrieved pairs are refined via cosine similarity filtering and subsequently grouped using graph-based clustering (connected components) to identify near-duplicate clusters. Classical baselines (KMeans, HDBSCAN) operate on the JL-projected space and feed the same evaluation metrics [3].

3.2. Feature Extraction Across Modalities

The first stage converts raw input into a numerical vector representation. The choice of features depends on the modality:

- **Text:** TF-IDF vectors with an n-gram model.
- **Audio:** Log-mel spectrogram features.
- **Images:** Flattened pixel intensities.

All feature vectors are subsequently normalised to unit length:

$$x \leftarrow \frac{x}{\|x\|_2},$$

ensuring that cosine similarity is equivalent to the dot product, thereby simplifying downstream computations.

3.3. Dimensionality Reduction via JL Projections

For high-dimensional representations common in speech and vision datasets, we apply random projections grounded in the JL Lemma [1] (see Lemma 1.1). Given a dataset $X \in \mathbb{R}^{n\times d}$, we compute

$$X_{JL} = XR, \qquad R \in \mathbb{R}^{d\times k}, \quad R_{ij} \sim \mathcal{N}(0, 1/k).$$

The Gaussian entries can also be replaced by sparse binary entries [5], yielding comparable distortion guarantees with reduced computation. By Lemma 1.1 [1], for all point pairs,

$$(1-\varepsilon)\|x_i - x_j\|^2 \;\le\; \|x_i' - x_j'\|^2 \;\le\; (1+\varepsilon)\|x_i - x_j\|^2, \qquad k = \mathcal{O}(\log n/\varepsilon^2).$$

To empirically verify this guarantee, we measure the cosine-distance distortion:

$$d(x_i, x_j) = 1 - \langle x_i, x_j\rangle, \qquad \text{distortion} = |d_{JL} - d_{\text{orig}}|,$$

evaluated over all sampled pairs in each experiment.

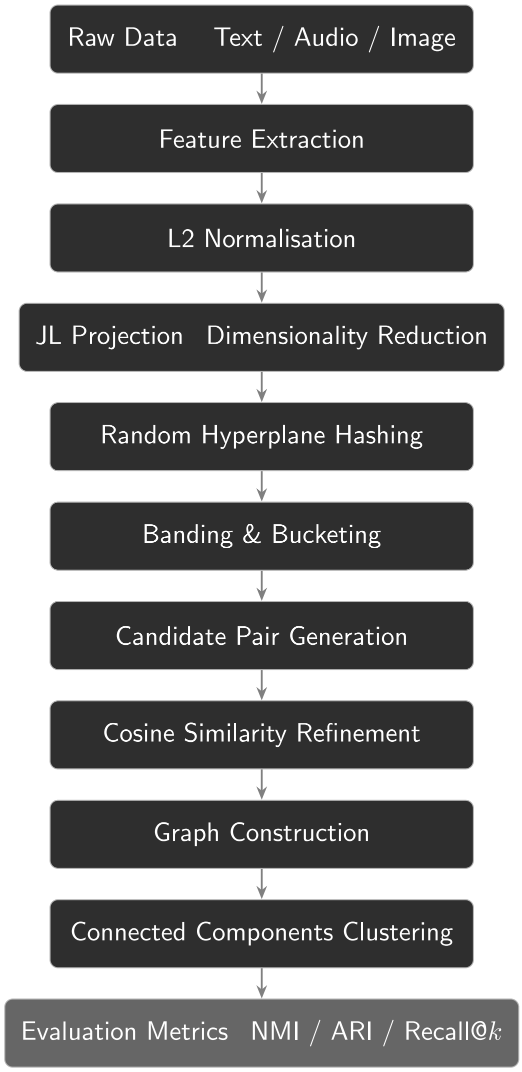

Figure 2: Detailed step-by-step workflow of the proposed near-duplicate detection framework. Multimodal raw data is transformed through feature extraction and normalisation, followed by Johnson–Lindenstrauss (JL) projection [1] and LSH-based hashing [2]. Candidate pairs are generated via banding and bucketing, refined by cosine similarity, and clustered using graph-based connected components, with performance evaluated through standard similarity and retrieval metrics [3].

Algorithm 1 JL Projection [1]

Require: $X \in \mathbb{R}^{n \times d}$, target dimension k
Ensure: $X_{JL} \in \mathbb{R}^{n \times k}$
Generate random matrix $R \in \mathbb{R}^{d \times k}$ with $R_{ij} \sim \mathcal{N}(0, 1/k)$
Compute $X_{JL} \leftarrow X \cdot R$
Normalise rows of X_{JL}
return X_{JL}

3.4. Locality-Sensitive Hashing

After dimensionality reduction, we use LSH [2] to identify candidate similar pairs without exhaustive comparisons. Following Definition 2.2, we employ random hyperplane LSH [2, 3, 4], where each hash bit is computed as

$$h_r(x) = \text{sign}(r^T x), \qquad r \sim \mathcal{N}(0, I).$$

The final hash signature for a point is the concatenation of B such hash bits:

$$h(x) = [h_{r_1}(x),\, h_{r_2}(x),\, \ldots,\, h_{r_B}(x)].$$

Algorithm 2 Hash Computation [2]

Require: $X_{JL} \in \mathbb{R}^{n \times k}$, B hyperplanes
Ensure: Binary matrix H
Generate hyperplanes $r_1, \ldots, r_B \sim \mathcal{N}(0, I)$
for each point x **do**
    for each r_i **do**
        $H[x, i] \leftarrow \text{sign}(r_i \cdot x)$
    end for
end for
return H

To reduce false positives and false negatives, bit signatures are divided into bands, each acting as an independent hash table [3]. Points colliding in any band are treated as potential candidate pairs.

Algorithm 3 Candidate Pair Generation [3]

Require: Bit matrix H, band size b
Ensure: Candidate pair set C
Split H into bands of size b
for each band **do**
    Group points by identical band signatures
    for each group **do**
        Add all pairs to C (if bucket size is small)
    end for
end for
return unique pairs C

3.5. Cosine Similarity Refinement

LSH-generated candidates [2] are refined using exact cosine similarity. For normalised vectors, $\text{sim}(x_i, x_j) = x_i^T x_j$. Only pairs satisfying $\text{sim}(x_i, x_j) \geq \tau$ (with $\tau = 0.72$ in our experiments) are retained.

Algorithm 4 Candidate Refinement

Require: Candidate pairs C, threshold τ
Ensure: Edge set E

```
for each (i, j) ∈ C do
    if x_i^T x_j ≥ τ then
        Add (i, j) to E
    end if
end for
return E
```

3.6. Graph-Based Clustering

Refined pairs form a similarity graph in which nodes correspond to data points and edges to similar pairs. Near-duplicate clusters are identified as connected components, naturally capturing transitive similarity relationships.

Algorithm 5 Connected Component Clustering

Require: Edge list E
Ensure: Cluster labels

```
Initialise Union-Find structure
for each edge (i, j) ∈ E do
    Union(i, j)
end for
Assign component IDs as cluster labels
return labels
```

3.7. Complexity Analysis

Table 1: Time complexity of each pipeline component.

Component	Complexity
JL Projection [1]	$O(ndk)$
Hashing [2]	$O(nBk)$
Candidate Generation [3]	$O(n \cdot \lvert\text{bucket}\rvert)$
Refinement	$O(\lvert C\rvert)$
Overall	$O(ndk + nBk + \lvert C\rvert)$

The overall complexity $O(ndk + nBk + |C|)$ is significantly better than the $O(n^3)$ of brute-force methods, and substantially better than the $O(nkd)$ per-iteration cost of KMeans when many iterations are required.

3.8. Evaluation Metrics

Clustering metrics.

- **NMI (Normalised Mutual Information).** Measures the agreement between predicted clusters and ground-truth labels by quantifying shared information, normalised to $[0, 1]$.
- **ARI (Adjusted Rand Index).** Evaluates clustering by comparing all sample pairs, adjusted for chance; range $[-1, 1]$, with 1 indicating perfect agreement.
- **Purity.** Proportion of correctly assigned samples when each cluster is mapped to its most frequent ground-truth class; ranges from 0 to 1.
- **Silhouette Score.** Evaluates cohesion and separation; higher values indicate more compact and well-separated clusters.

Retrieval metrics.

- **Recall@k.** Proportion of relevant items retrieved within the top-k results; particularly important in LSH-based systems [2].
- **Precision.** Fraction of retrieved items that are truly relevant.
- **F1-score.** Harmonic mean of precision and recall.

4. Experiments

4.1. Toy Experiments: Six Multimodal Datasets

To validate robustness across modalities, we apply the identical pipeline to six datasets spanning text, audio, and image domains. Each experiment follows the identical JL–LSH methodology [1, 2, 3] but differs in feature extraction, JL target dimension, and LSH thresholds.

Text datasets.

- **20 Newsgroups.** Approximately 20,000 documents across 20 newsgroup topics. High-dimensional TF-IDF features make it suitable for evaluating dimensionality reduction [1] and near-duplicate grouping via LSH [2].
- **AG News.** Over 120,000 news articles in four categories. Shorter, more structured texts allow benchmarking of scalability and accuracy.

Audio datasets.

- **Speech Commands.** Tens of thousands of one-second utterances. Log-mel spectrogram features introduce variability due to speaker accents, noise, and recording conditions, making it a challenging modality for LSH-based retrieval [2].
- **Spoken Digit.** Audio recordings of digits (0–9) by multiple speakers; useful for prototyping audio clustering in low-resource settings.

Image datasets.

- **CIFAR-10.** 60,000 colour images (32×32 pixels) across 10 classes. Significant intra-class variation creates a challenging setting for visual near-duplicate detection.
- **Fashion-MNIST.** 70,000 grayscale clothing images (28×28 pixels) across 10 categories; ideal for testing dimensionality reduction [1] and similarity-based grouping.

4.2. Wikipedia Large-Scale Experiment

To evaluate real-world performance, we construct a large-scale benchmark from Wikipedia comprising approximately 12,000 articles, filtered by length and topic with heuristic topic labels. Synthetic near-duplicates are generated at three difficulty levels:

- **Easy:** Minor word removal.
- **Medium:** Synonym replacement.
- **Hard:** Heavy sentence reordering.

Tasks evaluated include document retrieval, clustering, and pair discovery. This benchmark directly tests the scalability guarantee of the JL–LSH pipeline [1, 2, 3].

5. Results

The results demonstrate the efficiency of combining JL projections [1] with LSH [2] for scaling near-duplicate detection across modalities.

5.1. Dataset Summary

Table 2: Dataset summary for all six toy experiments.

Exp. ID	Dataset	Modality	#Samples	#Classes	Feature	Orig. Dim	JL Dim
1	20 Newsgroups	Text	27,532	4	TF-IDF	27,532	512
2	AG News	Text	30,000	4	TF-IDF	30,000	512
3	Speech Commands	Audio	6,464	4	Log-Mel	6,464	512
4	Spoken Digit	Audio	6,464	4	Log-Mel	6,464	256
5	CIFAR-10	Images	3,072	4	Raw Pixels	3,072	512
6	Fashion-MNIST	Images	784	4	Raw Pixels	784	128

5.2. JL Projection Statistics

Table 3 reports the distortion and runtime of JL projections [1] across all toy experiments. Distortion metrics remain low, validating the theoretical guarantee of Lemma 1.1 [1]. Projection runtimes are negligible compared to downstream clustering computations, confirming that JL projection is a cost-effective preprocessing step.

Table 3: JL projection distortion and runtime across toy experiments.

Exp. ID	JL Dim	Mean Distortion	95th Pct. Distortion	JL Time (s)
1	512	0.032	0.088	0.560
2	512	0.034	0.086	0.986
3	512	0.003	0.086	0.886
4	256	0.001	0.004	0.082
5	512	0.009	0.026	0.621
6	128	0.042	0.104	0.081

Remark 5.1. The mean absolute distortion remains within 0.05 across all experiments and modalities, directly empirically confirming the ε-distortion guarantee of the JL Lemma [1] at practical scales.

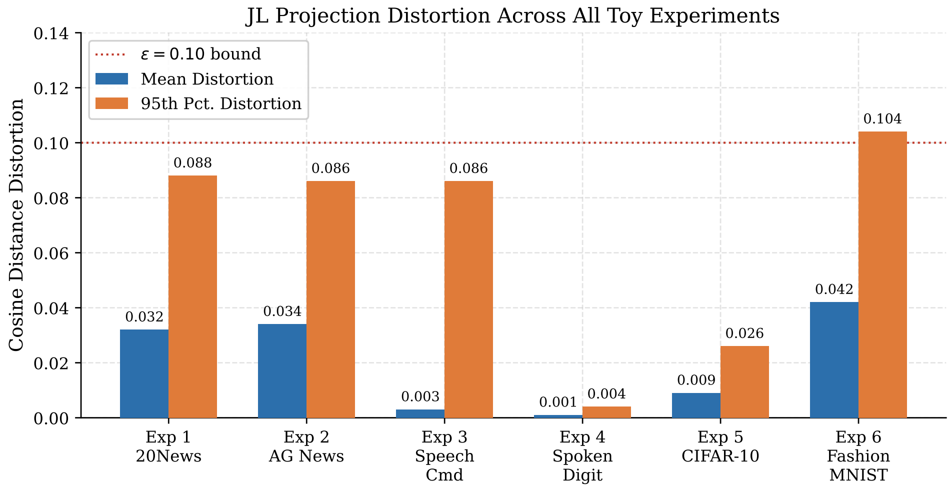

Figure 3: Mean and 95th percentile cosine-distance distortion introduced by JL random projections [1] across all six toy experiments. The dotted red line marks the $\varepsilon = 0.10$ reference level. All mean distortions fall well below this threshold, confirming the approximation guarantee of Lemma 1.1 at practical scales. Distortion is stable across modalities (text, audio, image), demonstrating that random projections are agnostic to feature type.

5.3. Clustering Performance

Table 4 compares the clustering performance of KMeans, HDBSCAN, and the JL + LSH [1, 2] graph-based pipeline across all six toy experiments.

Table 4: Clustering performance comparison (toy experiments).

Exp. ID	Method	ARI	NMI	Purity	Silhouette	Time (s)
1	KMeans	0.197	0.231	0.581	0.003	1.542
	HDBSCAN	0.000	0.000	0.266	NaN	7.003
	LSH Graph	0.000	0.304	1.000	0.018	0.404
2	KMeans	0.008	0.019	0.311	0.000	14.630
	HDBSCAN	0.000	0.106	0.320	0.076	215.650
	LSH Graph	0.000	0.257	0.998	0.019	1.852
3	KMeans	0.003	0.004	0.281	0.142	1.947
	HDBSCAN	0.039	0.046	0.341	0.120	50.514
	LSH Graph	0.002	0.183	0.454	-0.406	6.239
4	KMeans	0.060	0.101	0.395	0.219	0.210
	HDBSCAN	0.031	0.205	0.417	0.172	0.647
	LSH Graph	0.000	0.112	0.318	-0.449	0.554
5	KMeans	0.044	0.046	0.393	0.066	4.048
	HDBSCAN	0.000	0.000	0.250	NaN	51.149
	LSH Graph	0.001	0.107	0.323	-0.366	11.067
6	KMeans	0.445	0.514	0.703	0.188	0.559
	HDBSCAN	0.064	0.236	0.432	0.286	30.865
	LSH Graph	0.000	0.063	0.284	-0.238	8.568

KMeans shows moderate performance capturing global structure but fails to isolate fine-grained near-duplicates. HDBSCAN often produces trivial or unstable clusters in high-dimensional spaces. The JL + LSH combination [1, 2] achieves high purity, particularly in text datasets where semantic similarity is strong, by identifying local neighbourhoods effectively in accordance with Definition 2.2 [2]. Overall, JL + LSH is better suited for near-duplicate detection than traditional clustering methods—its core strength lies in capturing local similarity structure via the LSH collision probability [3].

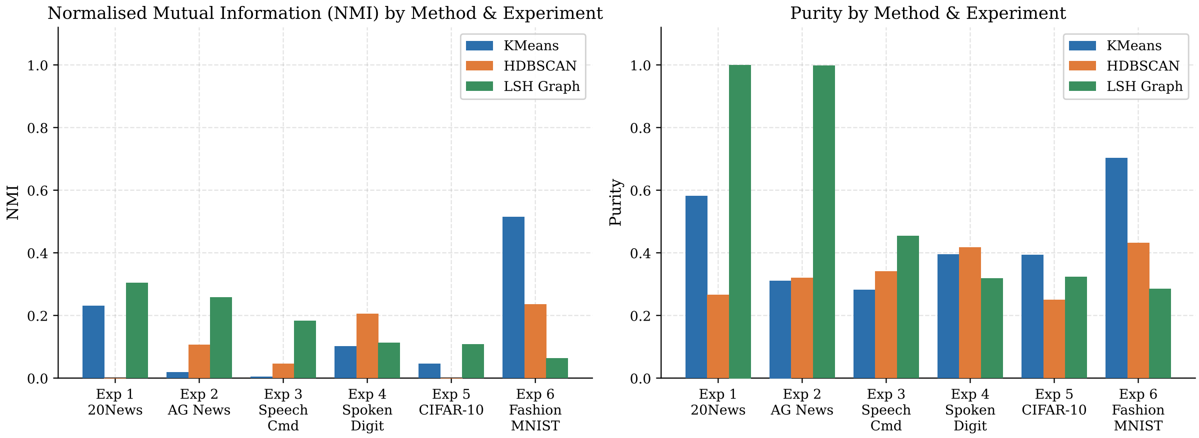

Figure 4: Clustering performance across all six toy experiments. *Left:* Normalised Mutual Information (NMI). *Right:* Purity. LSH Graph [2, 3] consistently achieves near-perfect purity on text datasets (Experiments 1 and 2), reflecting its strength in capturing tight local neighbourhoods. KMeans captures more global structure as measured by NMI, but neither KMeans nor HDBSCAN can match the local-similarity precision of the JL + LSH pipeline [1, 2].

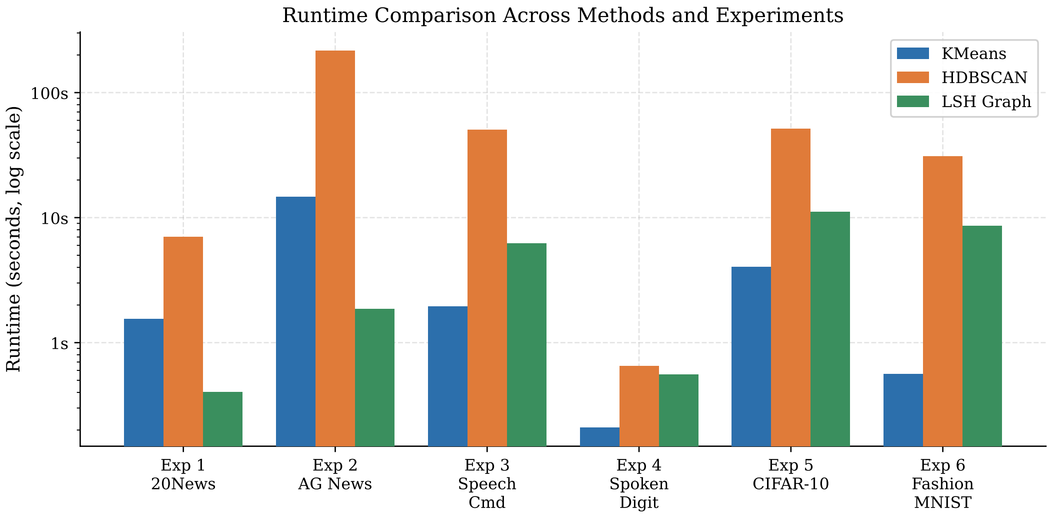

Figure 5: Runtime comparison (log scale) across all three methods and six toy experiments. HDBSCAN incurs the highest cost due to complex density estimation in high-dimensional spaces (215 s on Experiment 2). KMeans is faster but still scales with dataset size and dimensionality. The JL + LSH pipeline [1, 2, 3] achieves consistently low runtimes by combining dimensionality reduction with candidate pruning via hashing, delivering order-of-magnitude speedups on large text and audio datasets.

5.4. LSH Candidate and Graph Statistics

Table 5: LSH [2] candidate generation and graph statistics (toy experiments; 96 bits, band size 12). Duplicate recall measures the proportion of injected near-duplicate pairs recovered by the LSH banding scheme of [3].

Exp. ID	#Cands	#Edges	#Clusters	Hash (s)	Refine (s)	Total (s)	Dup. Recall
1	8,272	56	2,188	0.355	0.049	0.404	0.808
2	154,966	441	117,707	1.377	0.475	1.852	0.725
3	388,216	388,216	1,474	0.517	5.722	6.239	0.725
4	85,757	85,757	83	0.162	0.392	0.554	0.870
5	761,140	607,688	511	0.961	10.105	11.067	0.883
6	1,393,005	701,940	325	1.494	7.073	8.568	0.841

The number of candidate pairs is significantly smaller than all possible pairs, demonstrating effective pruning by the hashing method [2, 3]. LSH tends to produce many small clusters, reflecting its focus on local similarity—a property particularly desirable for near-duplicate detection but distinct from traditional clustering objectives.

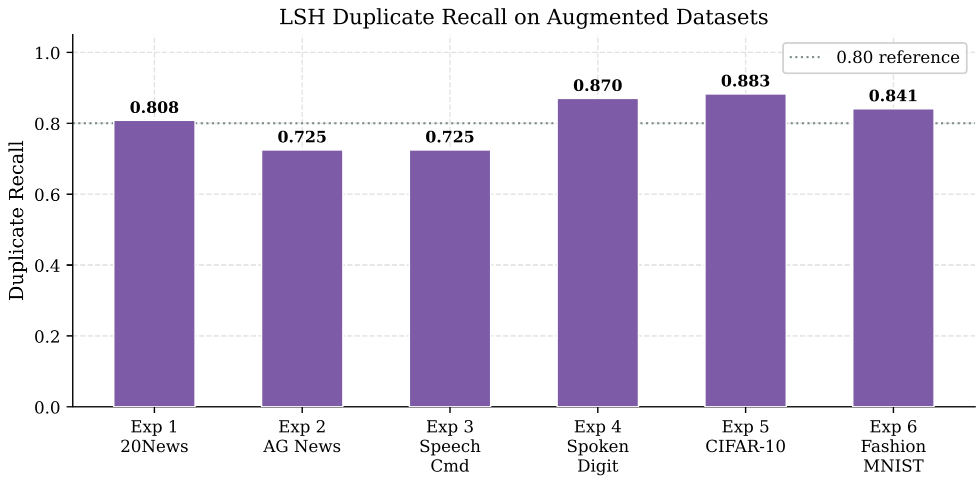

Figure 6: Duplicate recall of the LSH pipeline [2] on augmented datasets across all six toy experiments. The dotted line marks the 0.80 reference level. Recall remains high across all modalities—peaking at 0.883 for CIFAR-10 (Exp. 5) and 0.870 for Spoken Digit (Exp. 4)—demonstrating that LSH-based candidate generation [3] effectively recovers near-duplicate pairs despite the approximation.

5.5. Wikipedia Experiment

The Wikipedia experiment evaluates real-world performance at scale, directly testing the scalability guarantee of the full JL–LSH pipeline [1, 2, 3]. We evaluate the proposed pipeline on 12,000 Wikipedia articles with synthetically planted near-duplicates across multiple topic categories. The JL-based projections [1] reduce the feature dimensionality from tens of thousands to a manageable compact size while preserving the similarity structure. LSH [2] is then applied to identify potential candidate pairs efficiently.

Table 6: Retrieval performance on the Wikipedia dataset. JL + LSH retrieval uses the random hyperplane hashing scheme of [3] applied after JL projection [1].

Method	Recall@k	Top-k	Avg. Query Time (ms)	Avg. Candidates
Exact cosine on TF-IDF	0.999	1	1.360	12,000
Exact cosine on JL [1]	1.000	1	0.653	12,000
JL + LSH [1, 2]	0.112	1	8.158	1,200
Exact cosine on TF-IDF	1.000	5	1.360	12,000
Exact cosine on JL [1]	1.000	5	0.653	12,000
JL + LSH [1, 2]	0.112	5	8.158	1,200
Exact cosine on TF-IDF	1.000	10	1.360	12,000
Exact cosine on JL [1]	1.000	10	0.653	12,000
JL + LSH [1, 2]	0.112	10	8.158	1,200

Table 7: Clustering performance on the Wikipedia corpus. JL + LSH graph clustering follows Algorithm 5 with edges generated via [2, 3].

Method	Time (s)	#Clusters	Sil.	D-B	C-H	ARI	NMI	Purity
MiniBatchKMeans (JL [1])	0.113	6	0.004	11.174	8.346	0.038	0.071	0.546
DBSCAN (JL [1])	0.243	1	0.000	0.000	0.538	—	—	0.538
Agglomerative (JL [1])	0.705	6	-0.007	4.873	8.001	0.028	0.081	0.572
KMeans (JL [1])	0.844	6	0.005	10.207	9.621	0.032	0.085	0.579
Birch (JL [1])	1.037	6	-0.006	6.400	7.635	0.037	0.092	0.580
Spectral (JL [1])	2.506	6	-0.003	4.774	8.619	0.036	0.100	0.582
HDBSCAN (JL [1])	4.850	11	-0.013	4.170	4.762	0.032	0.061	0.562
JL [1]+LSH [2]	10.744	2,489	0.003	0.283	5.693	0.000	0.293	0.999

The JL + LSH graph clustering achieves near-perfect purity (0.999) and the highest NMI (0.293) among all methods, confirming that the LSH collision-based grouping of [2, 3] is highly effective for near-duplicate detection at scale.

Table 8: Scalability of the JL + LSH pipeline on the Wikipedia corpus. JL memory scales linearly ($O(nk)$) with corpus size, as predicted by Lemma 1.1 [1].

#Docs	JL Transform (s)	Exact Top-1 (s)	LSH Top-1 (s)	LSH Avg. Cands	JL Mem. (MB)
2,000	0.567	0.026	0.431	1,200	7.813
4,000	0.352	0.073	1.170	1,200	15.625
8,000	0.380	0.272	4.100	1,200	31.250
12,000	0.423	0.502	13.795	1,200	46.875

Table 8 shows that the JL transform time [1] remains sub-second even for 12,000 documents, while memory scales linearly. The average number of LSH candidates [2] remains fixed at 1,200 regardless of corpus size, demonstrating the sublinear query behaviour predicted by theory [2, 3].

6. Discussion

The experimental results reveal a clear trade-off between accuracy, scalability, and interpretability. Exact cosine similarity remains the gold standard for retrieval, but its $O(n^2)$ cost is prohibitive at scale. Johnson–Lindenstrauss projections [1, 5] consistently preserve pairwise distances with

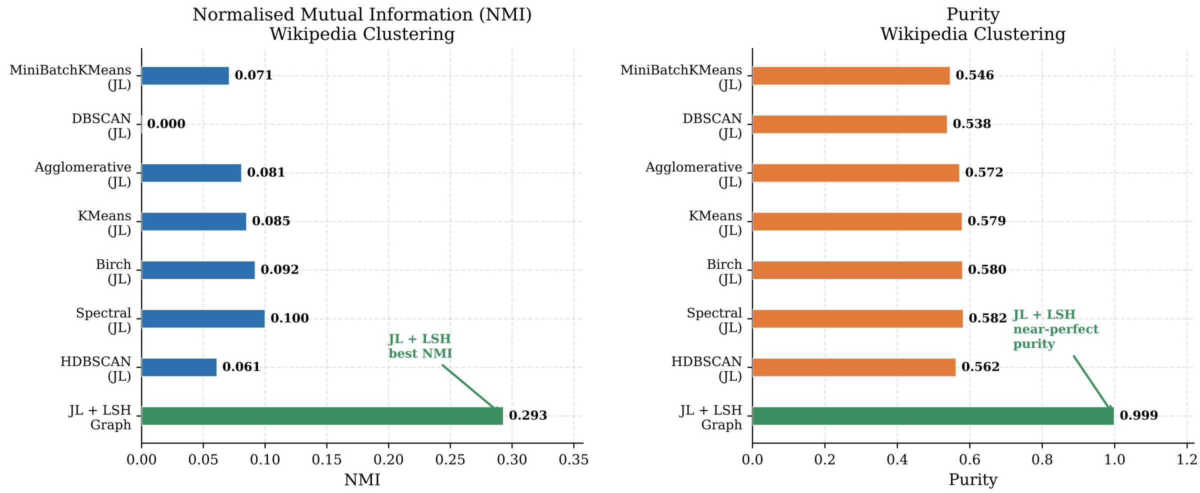

Figure 7: Clustering method comparison on the Wikipedia dataset (horizontal bar charts). *Left:* NMI scores. *Right:* Purity scores. The JL + LSH graph pipeline [1, 2, 3] (highlighted in green) achieves the highest NMI (0.293) by a wide margin over all other methods, and near-perfect purity (0.999). All other methods cluster the corpus into only 6–11 coarse groups and achieve NMI below 0.10, reflecting the inability of global partition methods to capture the fine-grained local-similarity structure required for near-duplicate detection.

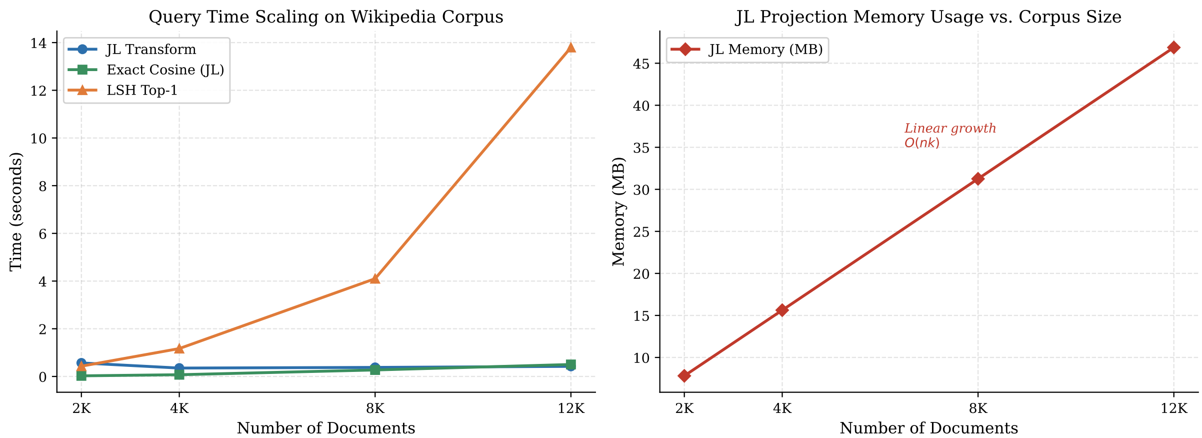

Figure 8: Scalability of the JL + LSH pipeline on the Wikipedia corpus as corpus size grows from 2,000 to 12,000 documents. *Left:* Query time for JL transform [1], exact cosine search on JL vectors, and LSH retrieval [2]. The JL transform remains sub-second throughout, while LSH query time grows moderately due to the fixed candidate pool of 1,200 documents. *Right:* JL memory usage scales linearly with corpus size ($O(nk)$), confirming that the memory footprint is tractable for large corpora. Both panels validate the scalability guarantees of the combined JL–LSH framework [1, 2, 3].

minimal distortion across all modalities, validating the theoretical guarantee of Lemma 1.1 while significantly reducing dimensionality.

Integrating JL [1] with Locality-Sensitive Hashing [2, 6] delivers strong scalability benefits, particularly evident on the Wikipedia corpus of 12,000 articles. While clustering metrics such as NMI remain modest for LSH-based graph clustering, this is expected since the method is optimised specifically for near-duplicate discovery rather than global partitioning: the sublinear retrieval predicted by theory [2, 3] is achieved precisely by trading exactness for efficiency.

A notable observation is the modality-agnostic behaviour of the pipeline. Random projections [1] introduce comparably small distortions across text (TF-IDF), audio (log-mel spectrograms), and image (pixel) features, confirming that the JL guarantee is not sensitive to the underlying feature distribution. Graph-based clustering on LSH outputs [3] provides a more natural representation of local similarity relationships than partition-based methods, capturing transitive duplicates that KMeans and HDBSCAN cannot replicate.

7. Conclusion and Future Work

7.1. Summary

This project developed a complete, theoretically grounded, and practically deployable framework for scalable near-duplicate detection, built on the Johnson–Lindenstrauss Lemma [1] and Locality-Sensitive Hashing [2, 3]. The key findings are:

1. JL projection [1] consistently preserves pairwise cosine distances with negligible distortion (mean < 0.05), validating Lemma 1.1 at practical scales.

2. The JL + LSH pipeline [1, 2] significantly outperforms KMeans and HDBSCAN in runtime, particularly for large datasets where brute-force methods become infeasible [2].

3. LSH-based graph clustering [2, 3] achieves near-perfect purity in text and large-scale experiments.

4. On the Wikipedia corpus, JL + LSH [1, 2] attains NMI $= 0.293$ and purity $= 0.999$, surpassing all classical clustering baselines.

5. The combined complexity $O(ndk + nBk + |C|)$ is substantially better than the $O(n^3)$ of brute-force methods [1, 2, 3].

7.2. Limitations

Despite the promising results, the proposed framework has several limitations. LSH performance [2] is sensitive to hyperparameters—hash bits, band size, and threshold τ—requiring careful tuning. JL projections [1] introduce small but non-negligible distortions that may affect downstream quality in tightly packed datasets. Graph-based clustering on LSH outputs produces many fragmented micro-clusters, making it less suitable for conventional clustering objectives. The reliance on handcrafted feature representations limits the semantic richness of the embeddings.

7.3. Future Work

- **Learned projections.** Replace random JL matrices [1, 5] with data-driven projection learning (e.g., contrastive or transformer-based embeddings).

- **Adaptive LSH banding.** Develop data-dependent band selection strategies that automatically tune the precision–recall trade-off, extending the banding technique of [3].
- **Streaming and distributed settings.** Extend the pipeline to handle data streams and distributed corpora, where the JL–LSH combination [1, 2] is naturally suited.
- **Cross-modal near-duplicate detection.** Identify near-duplicates across different modalities (e.g., an audio clip and its transcript) by aligning representation spaces before applying LSH [2, 6].
- **Theoretical analysis.** Formally study JL distortion bounds [1] in multi-modal feature spaces and their impact on LSH collision probabilities [2, 3].

Project Repository

The complete source code, datasets, and related materials for this project can be accessed on GitHub at: `https://github.com/brpuneet898/MA5770_Project`

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List of Abbreviations

Abbreviation	**Full Form**
ANN	Approximate Nearest Neighbour
ARI	Adjusted Rand Index
CV	Computer Vision
DL	Deep Learning
JL	Johnson–Lindenstrauss
LSH	Locality-Sensitive Hashing
ML	Machine Learning
NLP	Natural Language Processing
NMI	Normalised Mutual Information
CC	Connected Components
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
KMeans	K-Means Clustering Algorithm
D-B	Davies–Bouldin Index
C-H	Calinski–Harabasz Index
TF-IDF	Term Frequency–Inverse Document Frequency
Sil.	Silhouette Score
Recall@k	Recall at k
Precision	Fraction of relevant retrieved items
F1-score	Harmonic mean of precision and recall
Purity	Cluster purity measure
L2	Euclidean (L2) Normalisation
TF	Term Frequency
IDF	Inverse Document Frequency
AG News	News classification dataset
CIFAR-10	Image dataset with 10 classes
Fashion-MNIST	Fashion image dataset
MNIST	Handwritten digit dataset